NAMIBIA UNIVERSITY
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HCI711S LAB 1

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Fingerprint Biometric technique

Widely accepted and familiar – Most people are comfortable using fingerprint recognition.

Cost-effective – It is generally cheaper to implement than other biometric methods. Advantages for small businesses

Easy to use – Simple and quick for users to scan their fingerprint.

Data collection

Ethical protocols were followed by first obtaining explicit permission from participants, who were informed their fingerprints would serve solely for academic purposes. To safeguard privacy, the biometric data was employed only for the demonstration and not retained. Additionally, anonymity was ensured by keeping the samples free of any identifying information like names.

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ETHICAL DATA COLLECTION AND CONSENT FORM

Project Title: Biometric Fingerprint Authentication System

This project aims to design and demonstrate a biometric fingerprint authentication system prototype for academic purposes, showing how fingerprints can be used to identify and authenticate users. Participants may provide a fingerprint scan and a basic identification label for testing the prototype. The collected data will only be used for testing the biometric system, demonstrating fingerprint recognition, and for academic evaluation of the project. Fingerprint data will be used only during the testing phase, stored securely, and deleted after the project evaluation. Participants will remain anonymous in reports or presentations. Participation is completely voluntary, and participants may refuse, withdraw at any time, or request deletion of their data without any consequences. By signing the form, participants confirm that they understand the project and agree to provide their fingerprint data for testing purposes.

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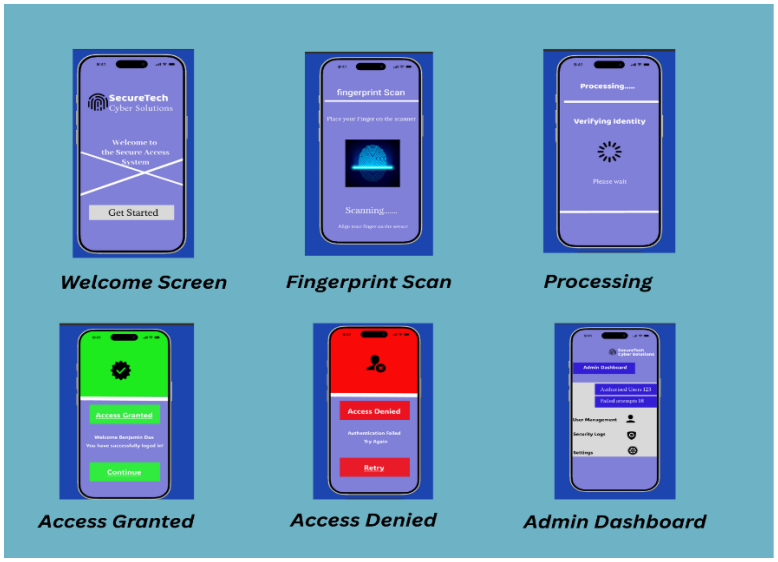
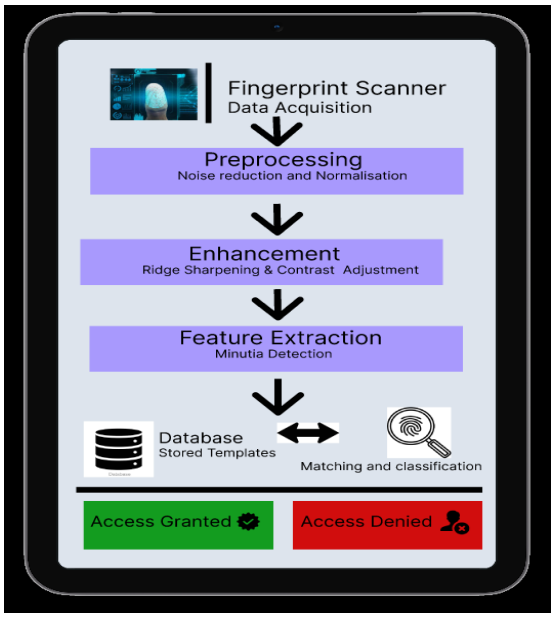
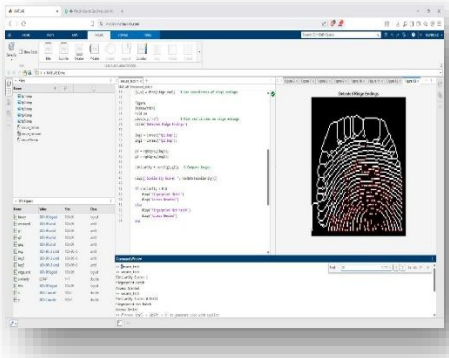
The prototype demonstrates a secure and functional fingerprint biometric authentication system that incorporates enhancement, preprocessing, feature extraction, classification, and surety mechanisms, while maintaining usability and corporate security standards. The system was developed using the Agile Software Development methodology, where: Requirements were defined, Screens were iteratively designed and Security features were refined.

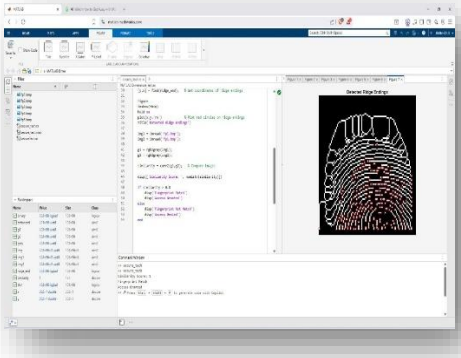


The steps involved in **fingerprint capture**, **processing, matching, and authentication** are depicted in the **block diagram**, which depicts the biometric fingerprint authentication system's workflow. A crucial system function is represented by each block.
These tasks are really carried out using **a Biometric Software Development Kit (SDK**). The SDK offers the tools and methods needed to process fingerprint photos, identify distinctive fingerprint characteristics, and compare them to database-stored fingerprint templates.
As a result, **the SDK handles the technical tasks inside each step of the process, while the block diagram depicts the overall system structure**. When taken as a whole, they show how the fingerprint authentication system works from user input to access determination.



The MATLAB simulation was used to analyse fingerprint images by extracting ridge ending features (minutiae) and comparing two fingerprint samples within a biometric authentication process. The system followed several key stages, including preprocessing, image enhancement, feature extraction, and matching, which represent the standard workflow of biometric recognition systems. MATLAB image processing techniques detected ridge endings and displayed them as red points on the skeletonized fingerprint image, confirming that the ridge patterns were accurately identified. The program then calculated a similarity score when comparing two fingerprint images, where a value of **1** indicated a strong match and resulted in the messages **"Fingerprint Match"** and **"Access Granted."** This simulation confirms that the prototype can identify important fingerprint characteristics, compare biometric data effectively, and make an access decision based on the calculated similarity.





References.

RecogTech. (n.d.). *5 common biometric techniques compared*. RecogTech. https://recogtech.com/en/insights-en/5-common-biometric-techniques-compared/

TPXimpact. (n.d.). *Ethical considerations for data collection | Ethical data collection*. Knowledge Hub. Retrieved March 7, 2026, from https://www.tpximpact.com/knowledge-hub/insights/ethical-data-collection